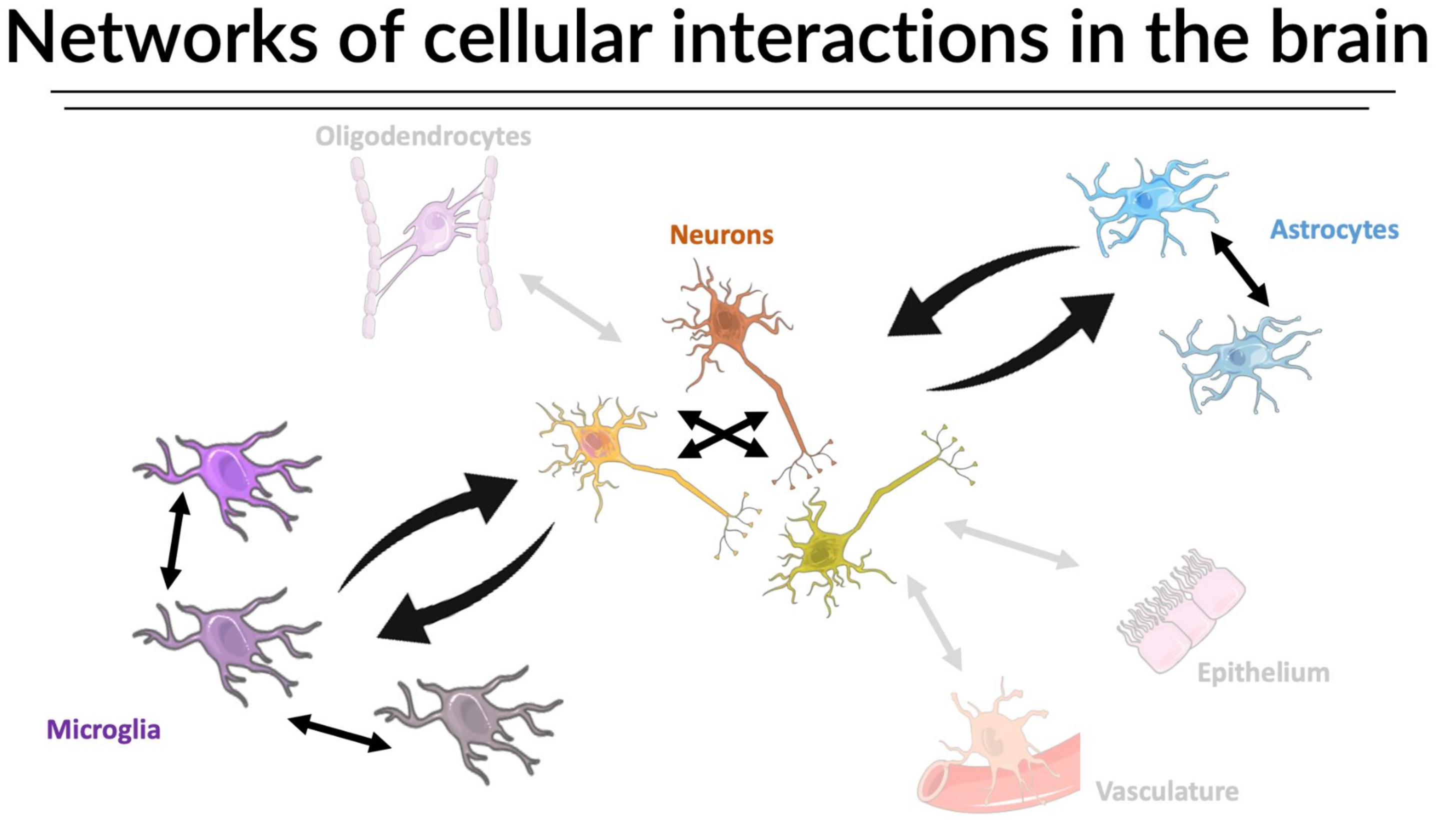
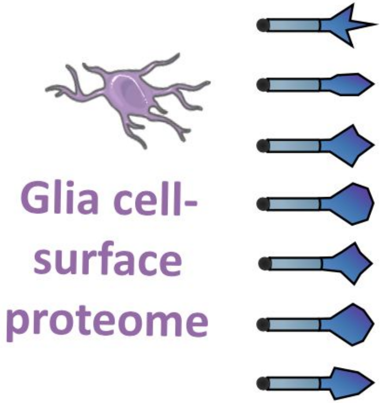
High-throughput mapping of brain cell interactions

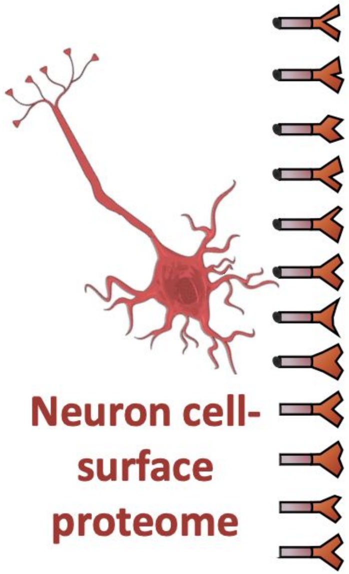
Background: interaction screen design



Background: interaction screen design

1. Define receptor repertoires of human brain neurons and glia

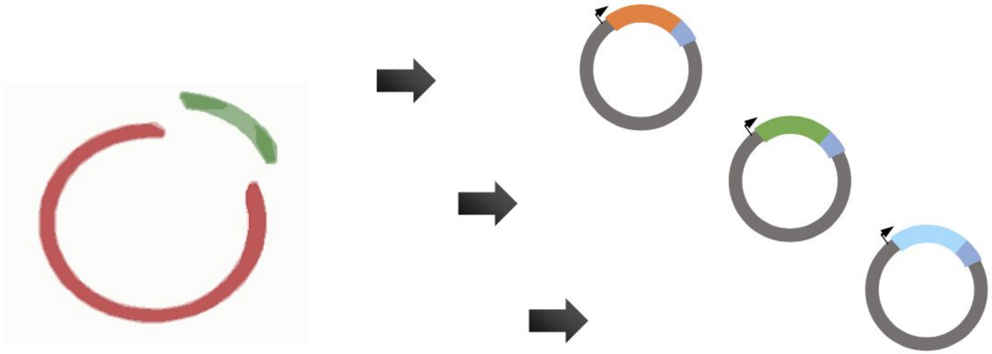




Background: interaction screen design

1. Define receptor repertoires of human brain neurons and glia

2. Generate recombinant protein libraries of the receptors

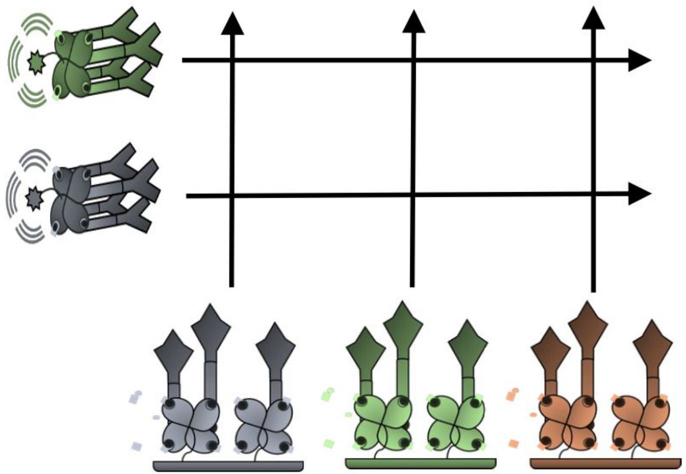


Background: interaction screen design

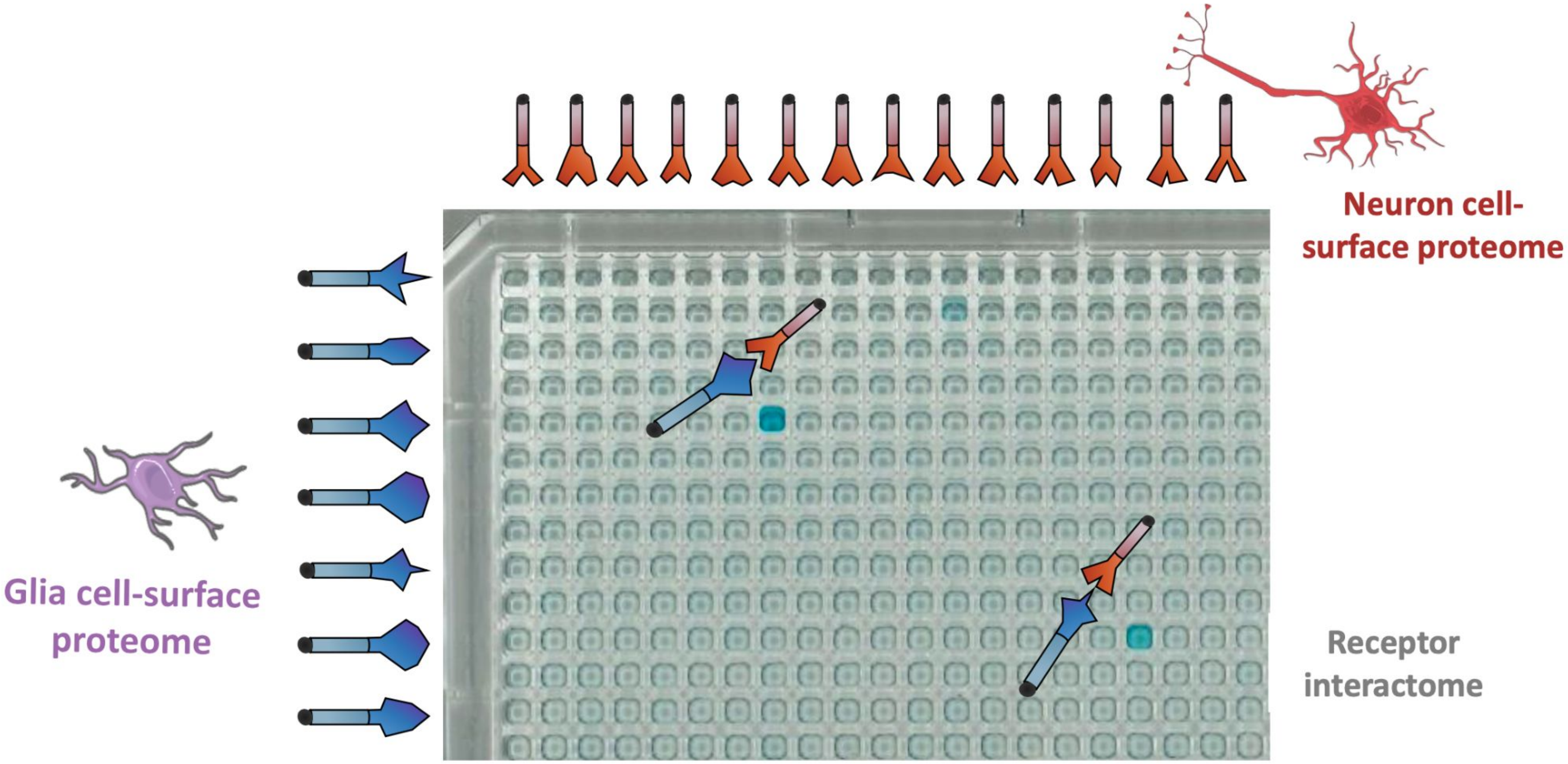
1. Define receptor repertoires of human brain neurons and glia

2. Generate recombinant protein libraries of the receptors

3. Screen all possible pairings of the receptors



Background: interaction screen example



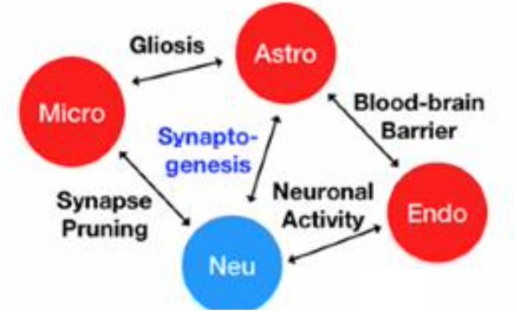
Background: principal focus of the rotation

1. Define receptor repertoires of human brain neurons and glia

2. Generate recombinant protein libraries of the receptors

3. Screen all possible pairings of the receptors

4. Analyze the interactome – known interactions & new discoveries



Objectives

(1) General interactome trends (Prefrontal cortex)

(2) PPI expression in different contexts (PFC, M1)

(3) Changes in patterns of CCIs across different disease states (AD, ALS)

(4) Specific PPIs of interest

Methods

- **Cell-cell interactions are assumed on the basis of expression shared between two cell types**
- **I considered three independent methods of comparing expression**

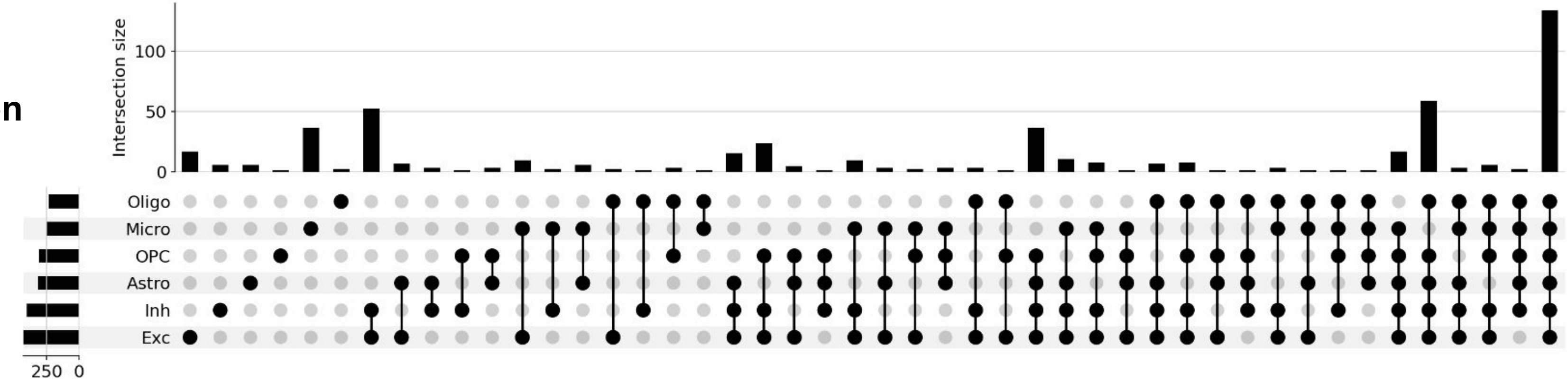
(1) **Raw expression across an entire cluster:** Proportion of a given cell cluster exhibiting >=1 UMI count, static threshold of >5% used as a filter

(2) **Gaussian mixture model:** Using a GMM to filter cells showing marginal expression - i.e. find the first “dip” in a multimodal (normalized) expression distribution across all clusters.

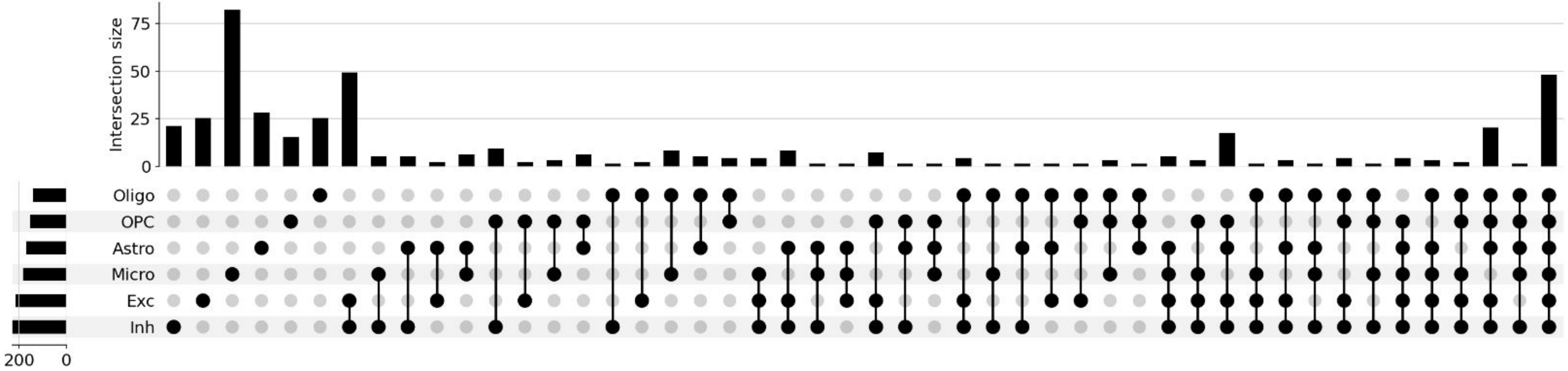
(3) **IQR:** Gene is only considered expressed in cells with a normalized count >Q1 across all clusters.

General interactome trends

Proportion

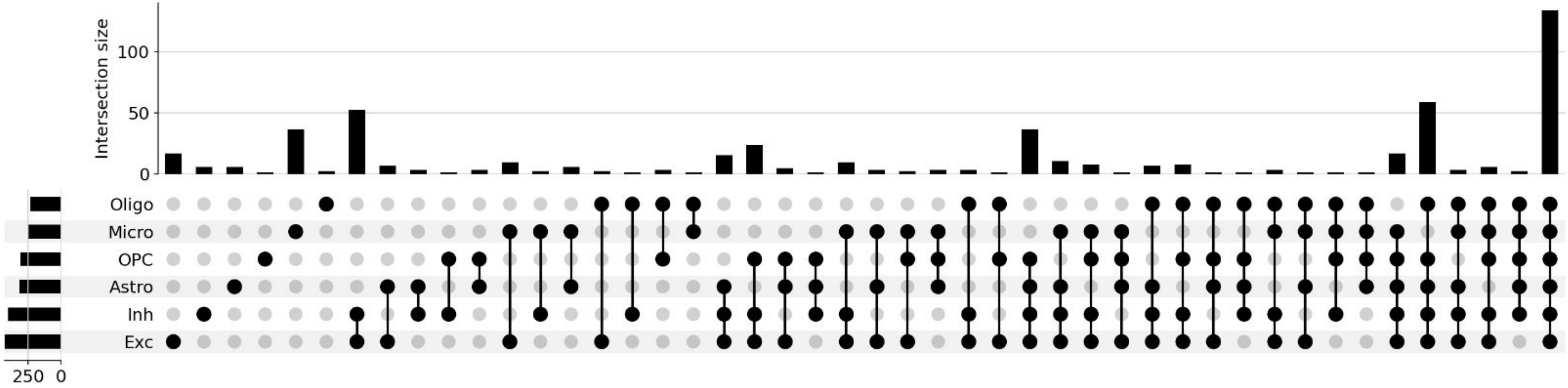


GMM



General interactome trends

Proportion



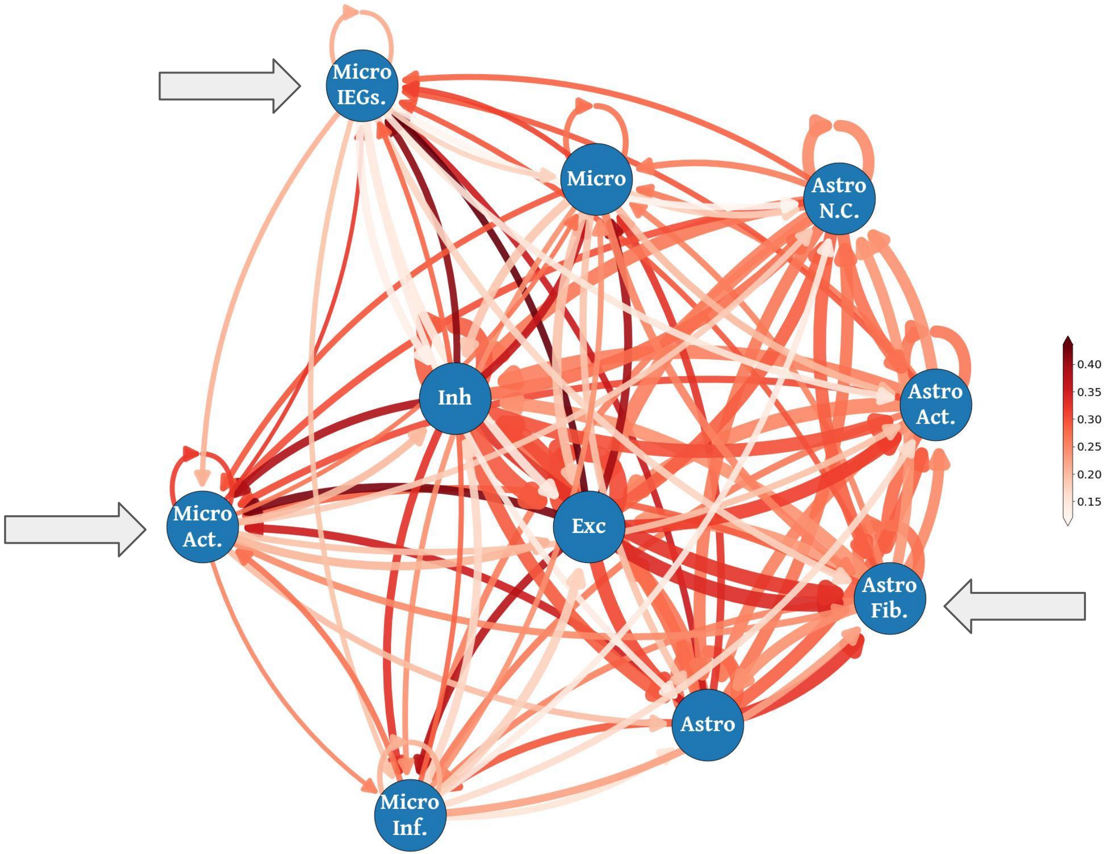
Screened: 517/523 (98.8%)*
Curated gene list: 200/201 (99.5%)*

General interactome trends

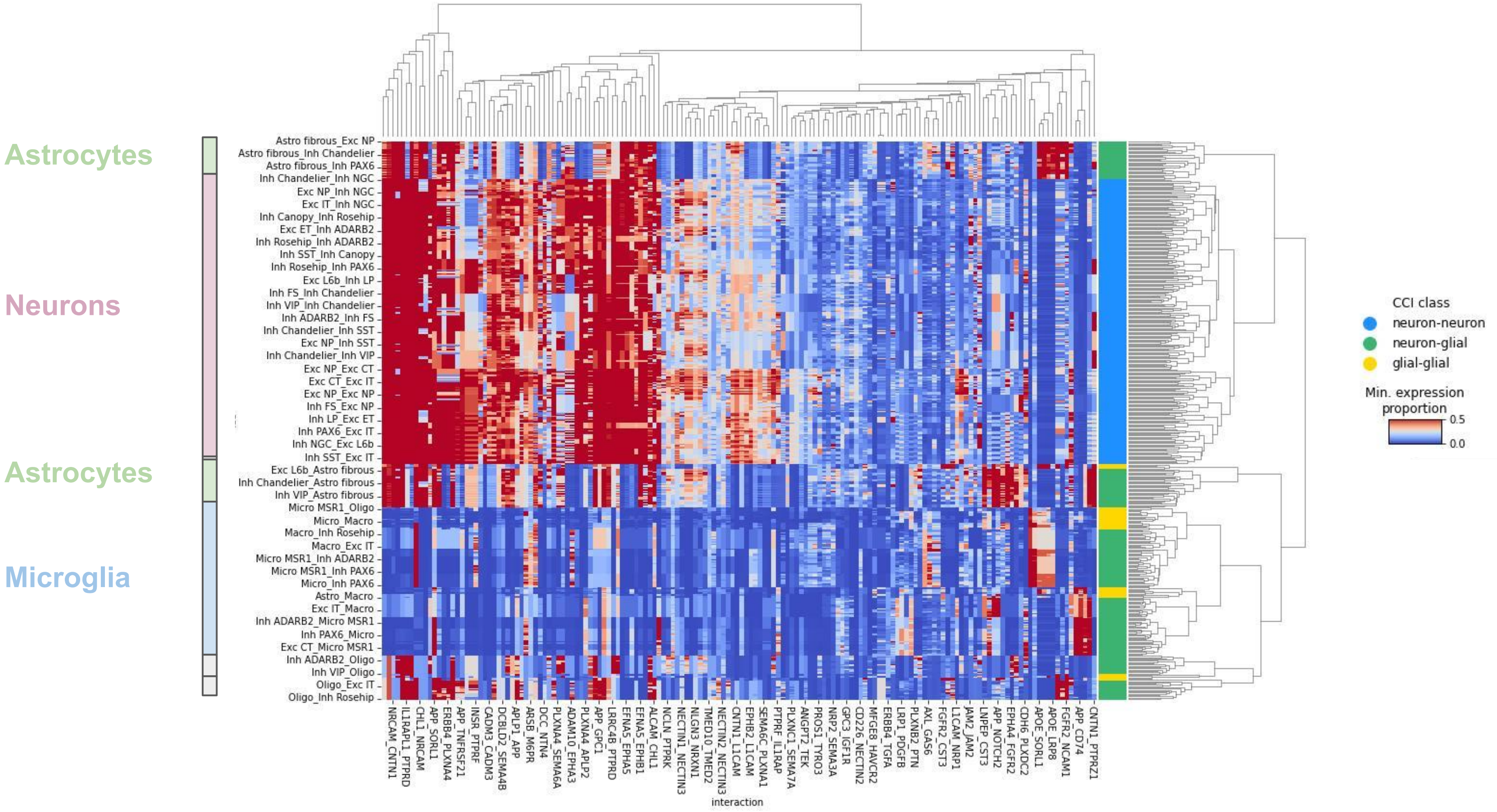
- Least novel interaction between neurons

- Glial - neuron CCIs have a higher proportion of novel PPIs

- Subtypes have higher proportion of novel interactions (specifically glia)

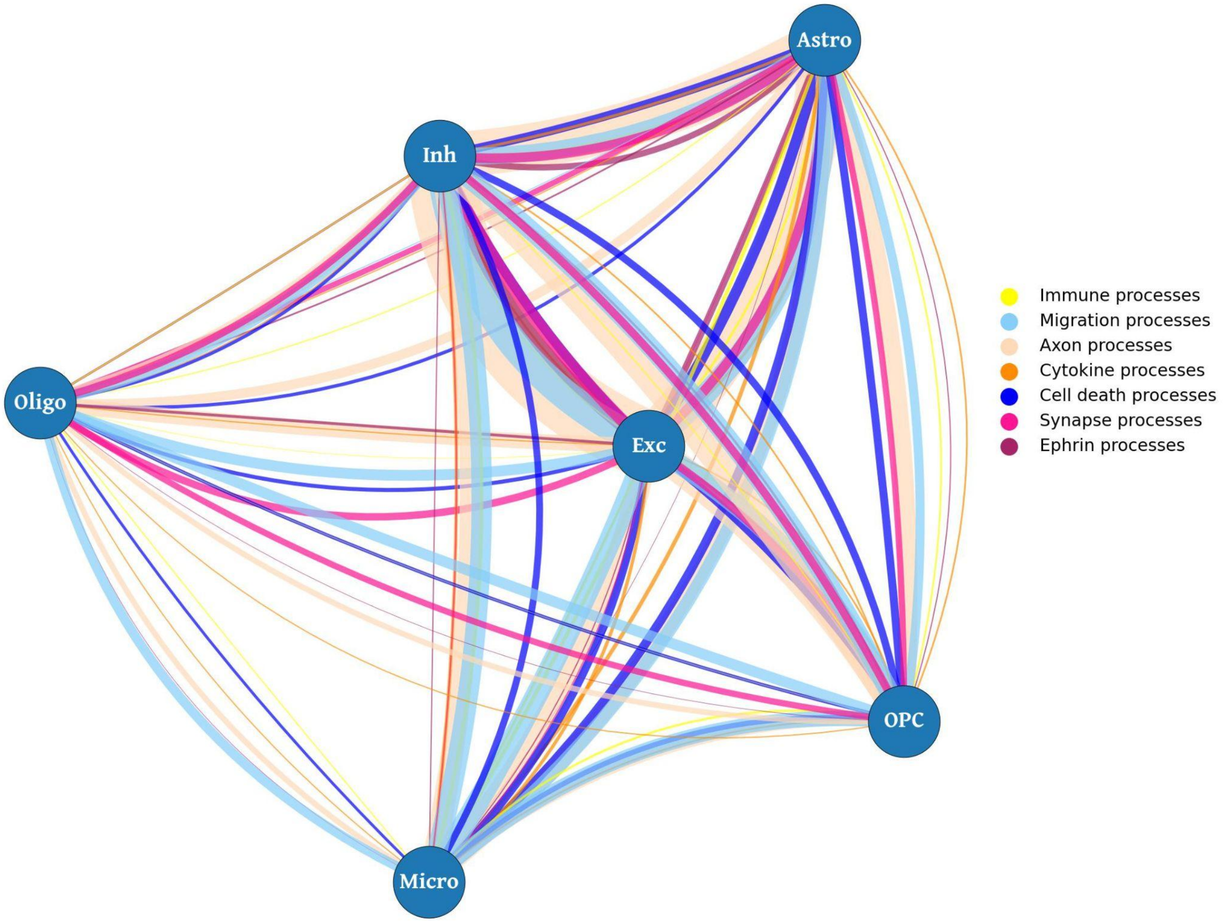


General interactome trends

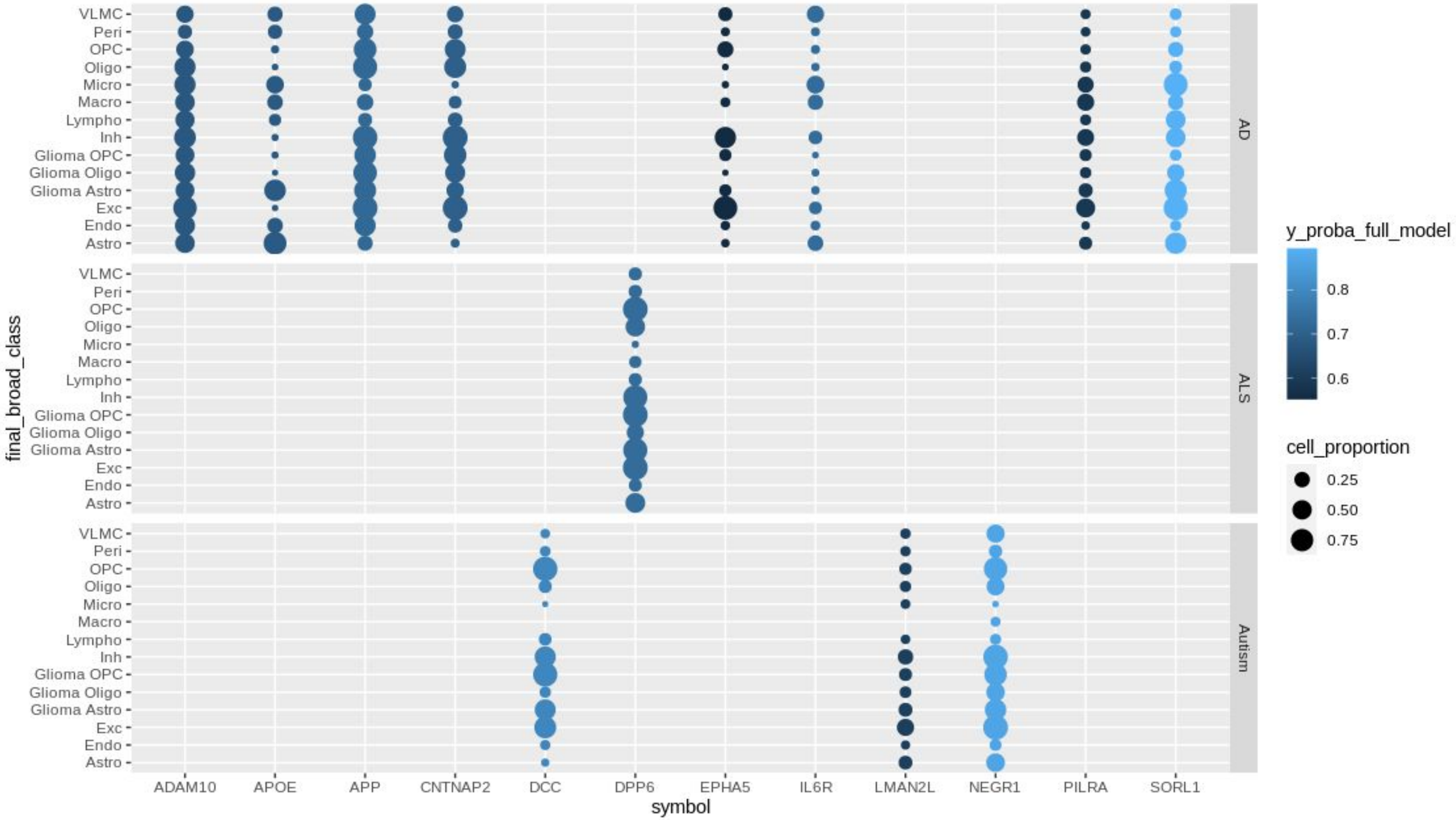


General interactome trends

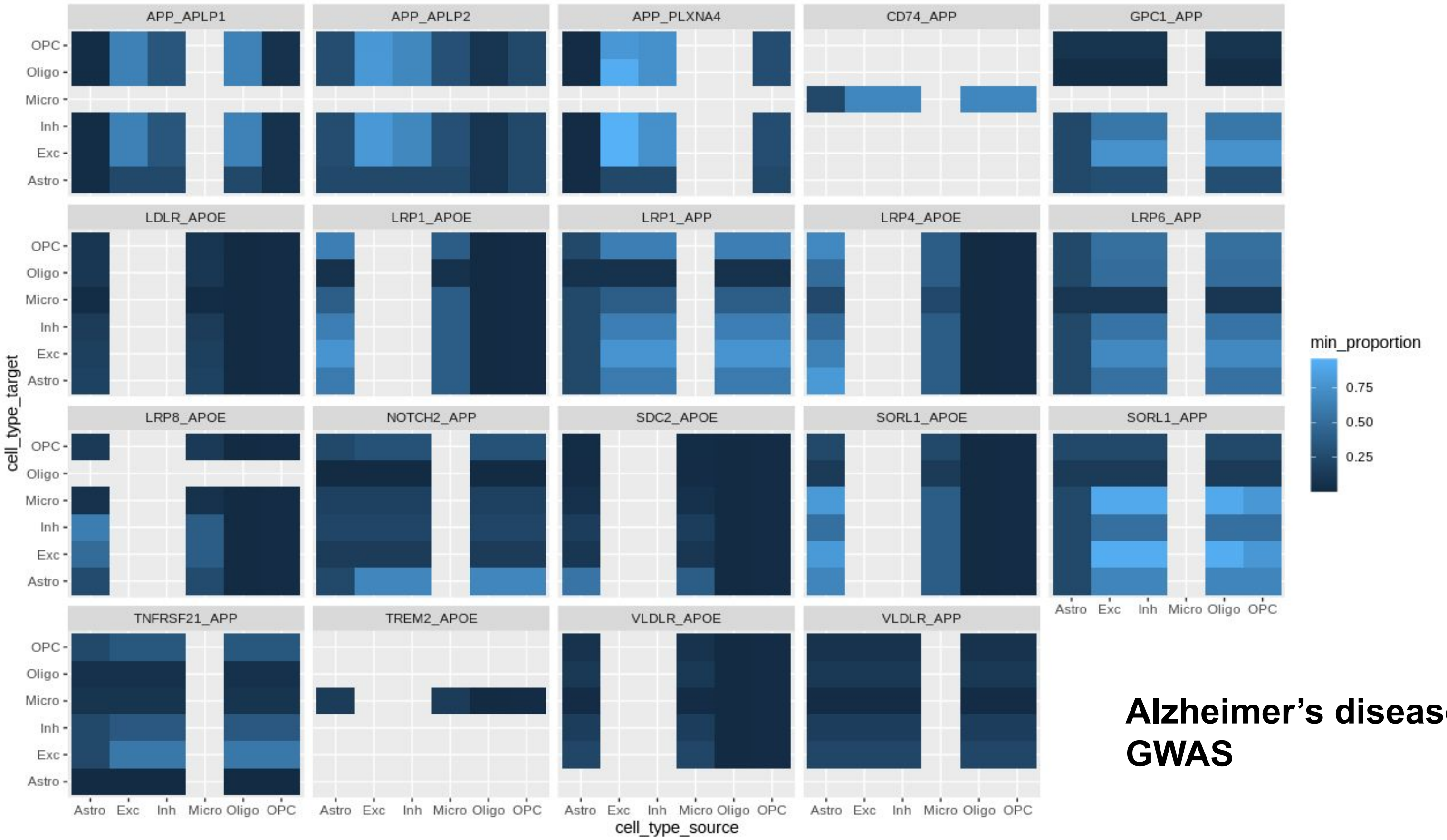
- Annotated PPI by shared GO terms and semantic similarity
- For visualization, added them to manually curated process categories
- Would be more interesting outside of a broad class cell type label



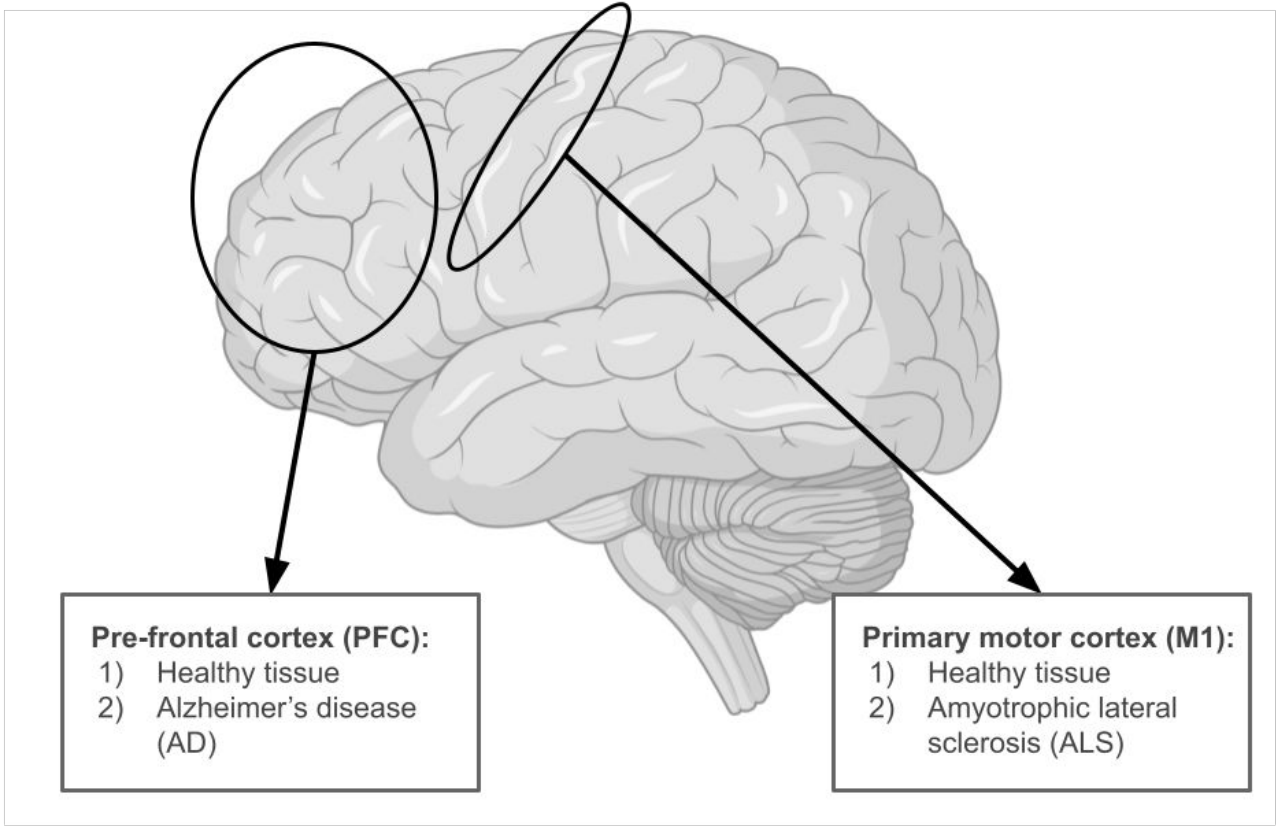
Assessment of pathological relevance



Assessment of pathological relevance



Alzheimer’s disease GWAS



Pre-frontal cortex (PFC):
1) Healthy tissue
2) Alzheimer's disease (AD)
Primary motor cortex (M1):
1) Healthy tissue
2) Amyotrophic lateral sclerosis (ALS)

Methods

- **Differential interaction activity between healthy and diseased samples was investigated using two complementary analyses**

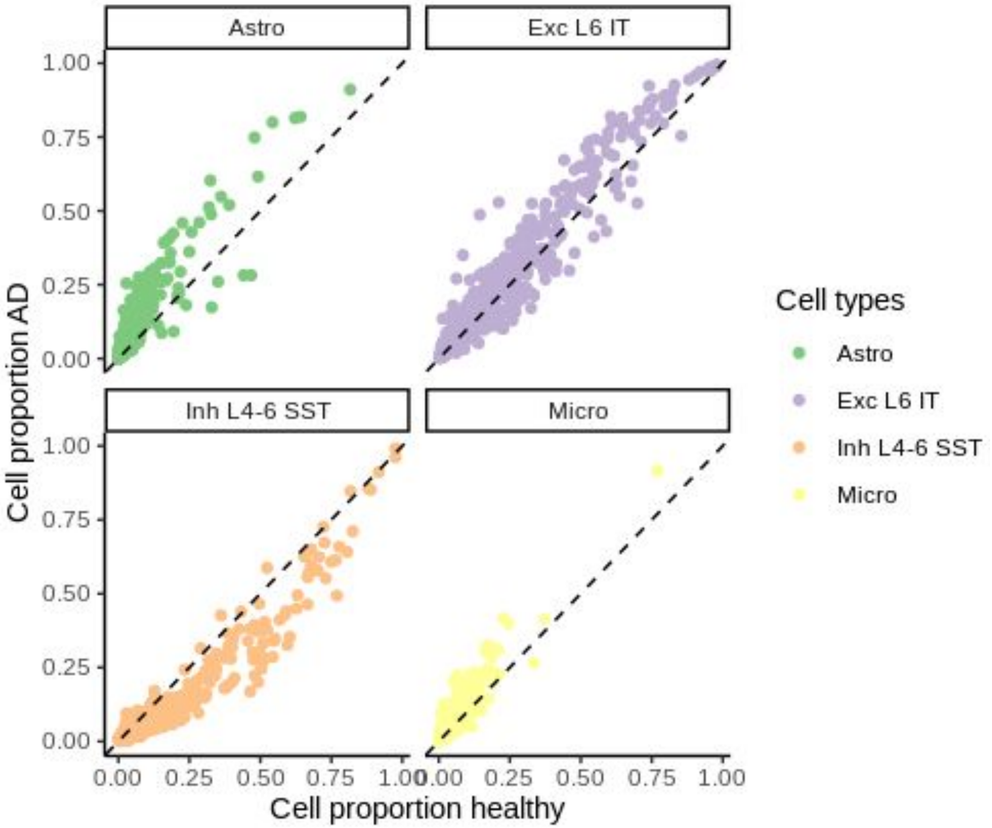
(1) Differential proportion of cells expressing a gene in a given cluster:

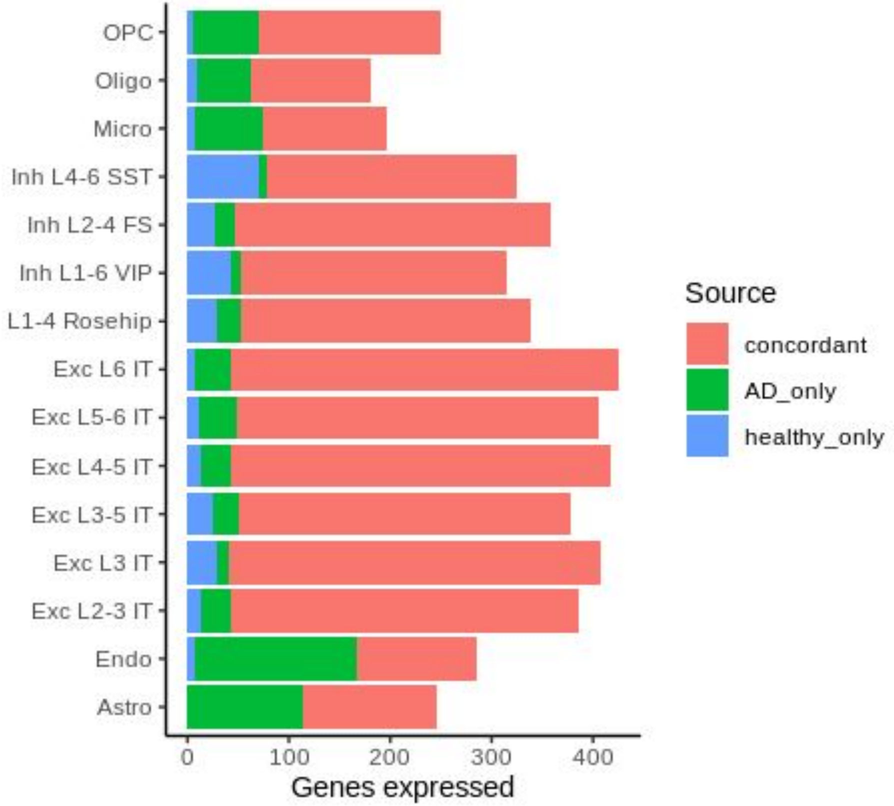
- Expression proportions were calculated for each sample in each cluster
- Differences between healthy and diseased cells were tested using a binomial regression model

(2) Differential gene expression of pseudobulked counts for each cluster:

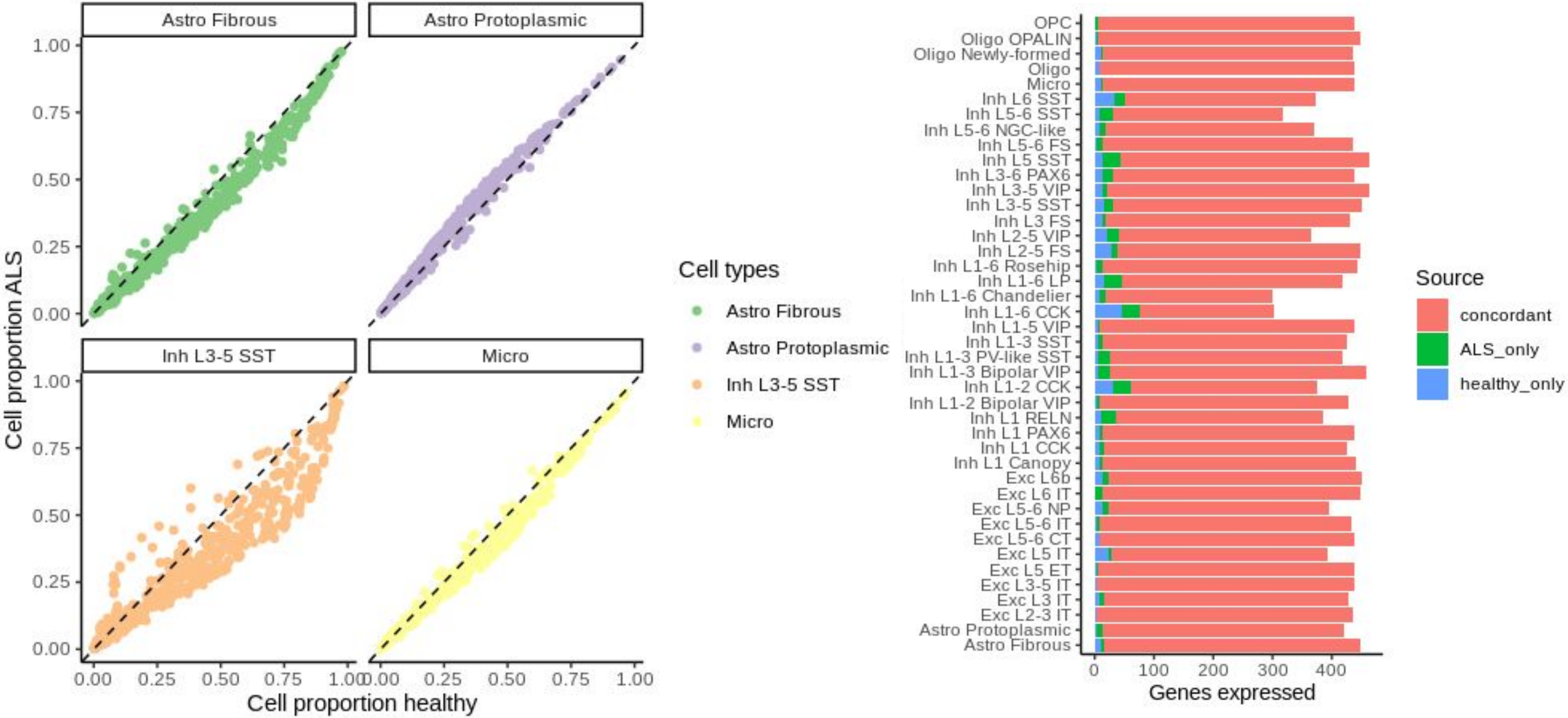
- Cells were pseudobulked on the basis of sample and cluster label
- These counts were evaluated using edgeR

Screen expression PFC healthy vs. AD

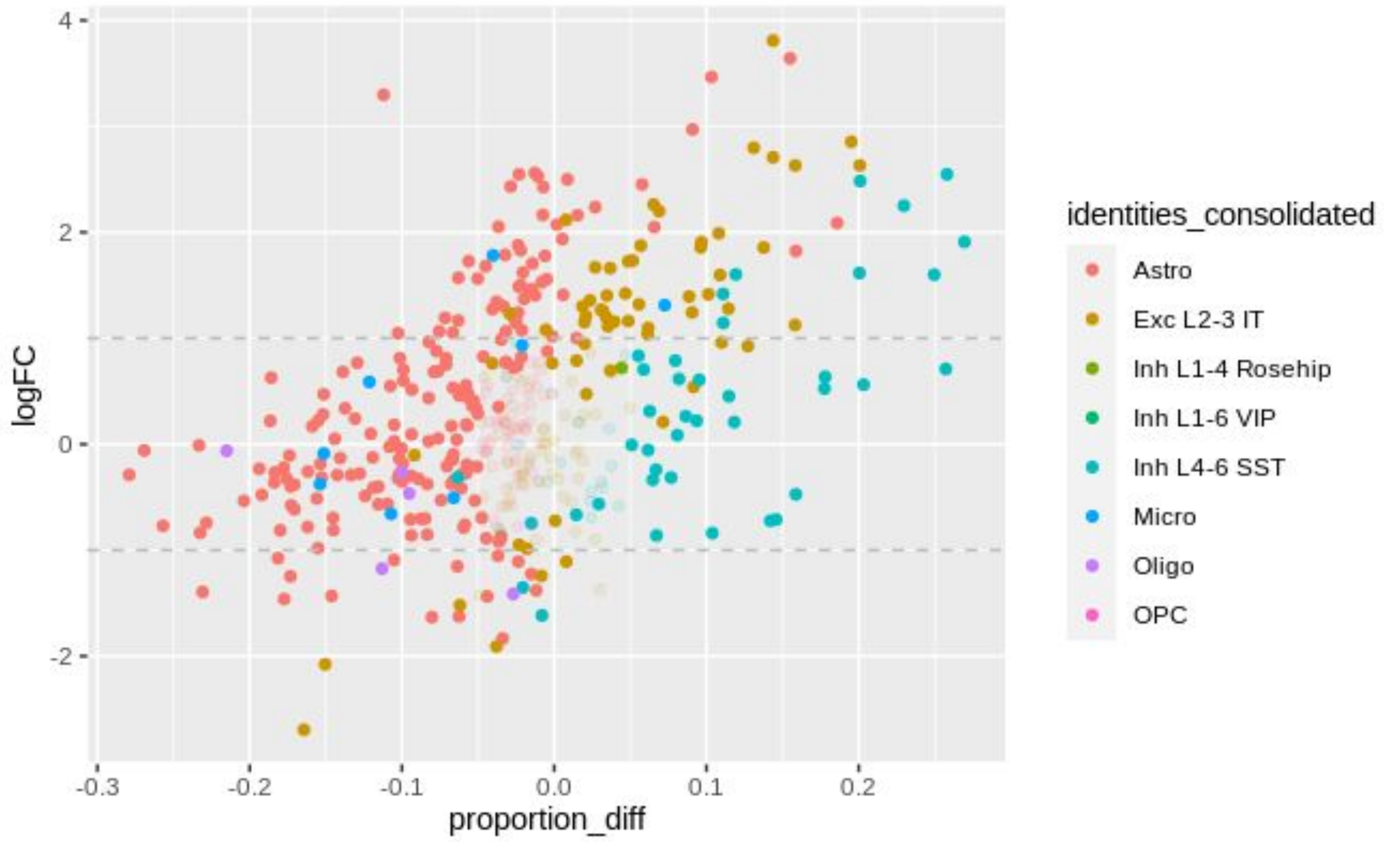




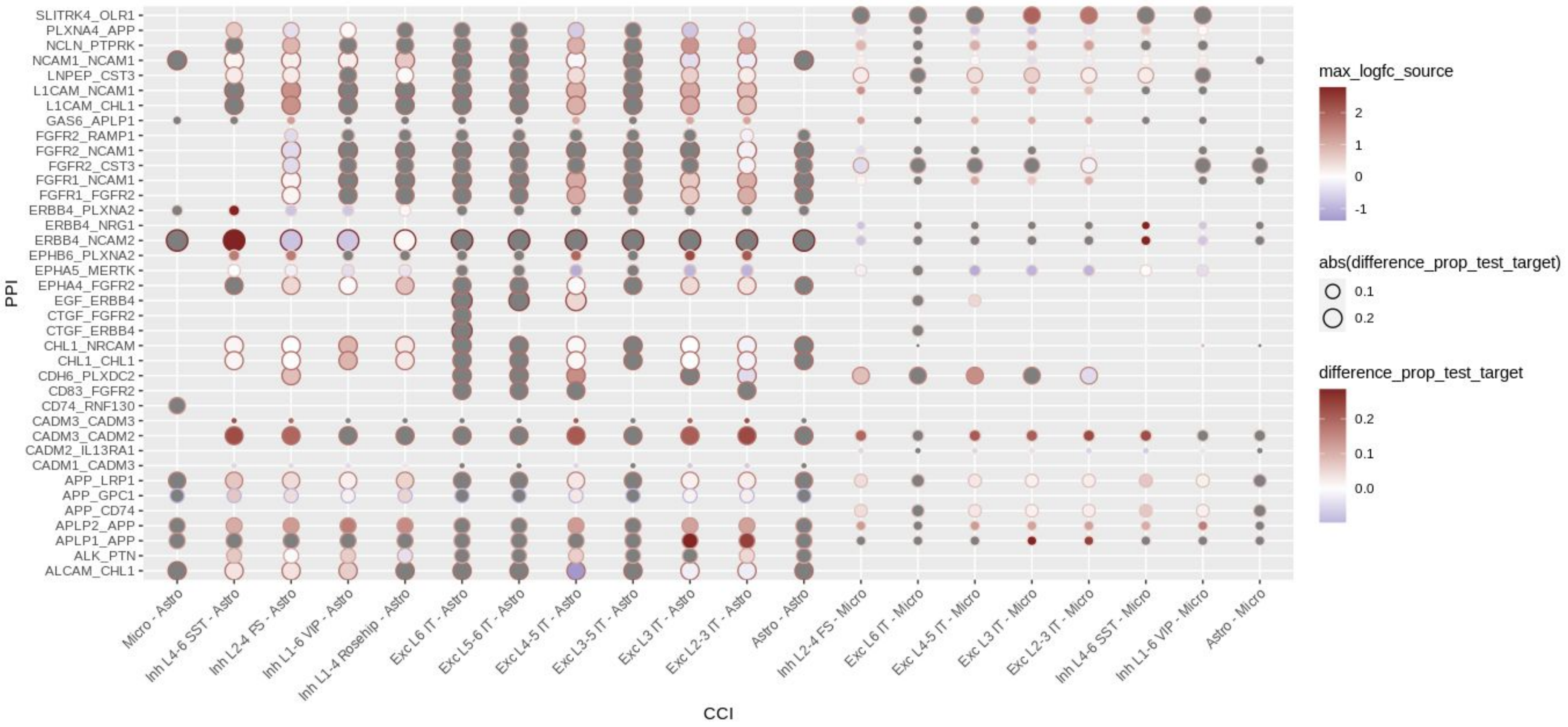
Screen expression MC1 healthy vs. ALS



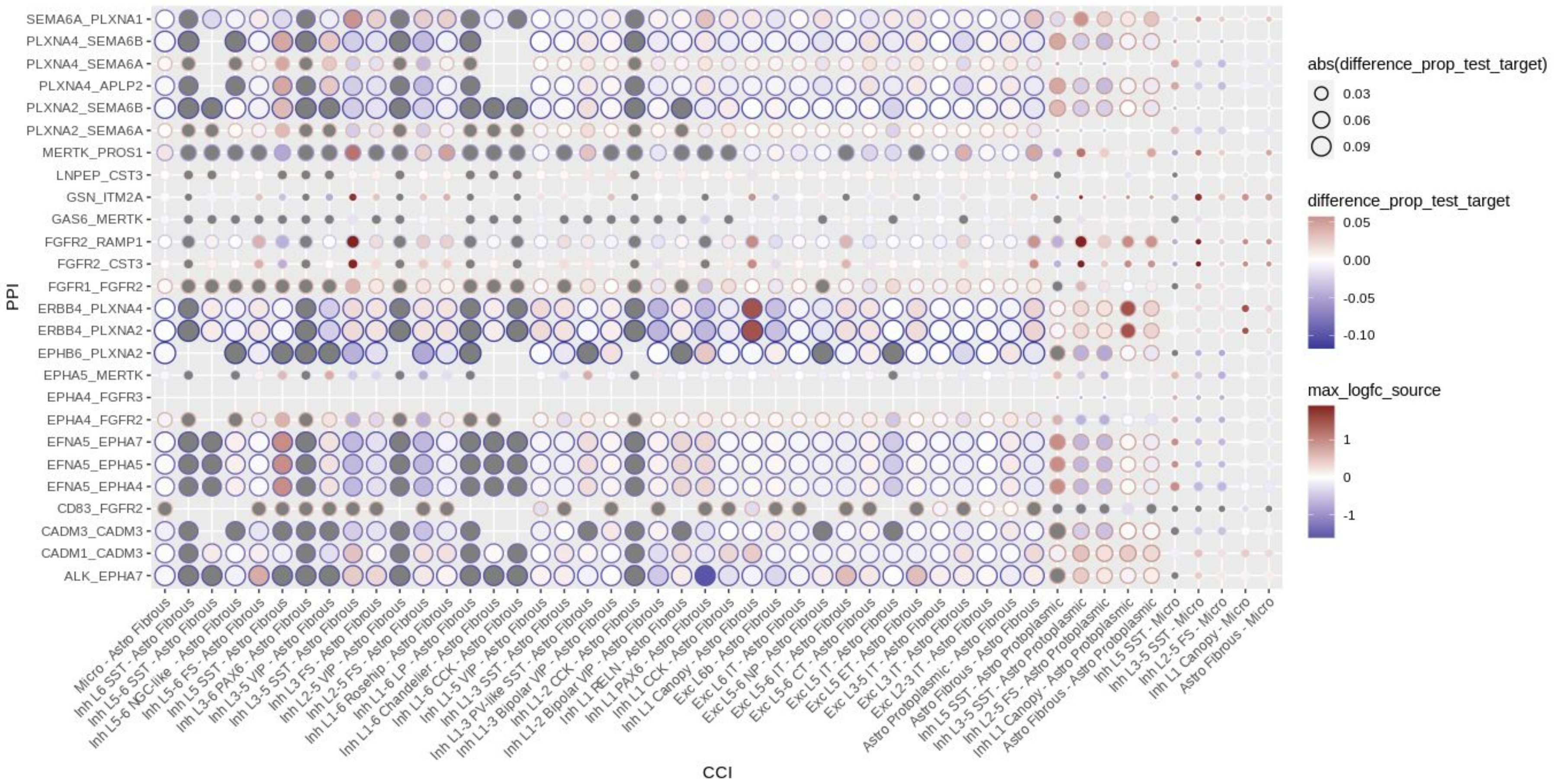
Methods: DEG, DEP example



Investigating logFC and differential proportions

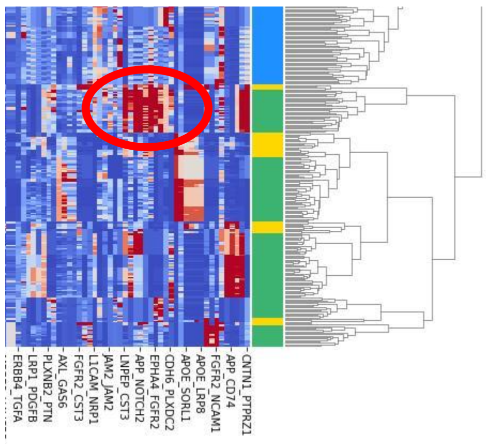


Investigating logFC and differential proportions



Planned and future directions

- **Follow up on cell-type specific interactions (i.e. heatmap clusters) and see whether they're modulated by in different contexts**



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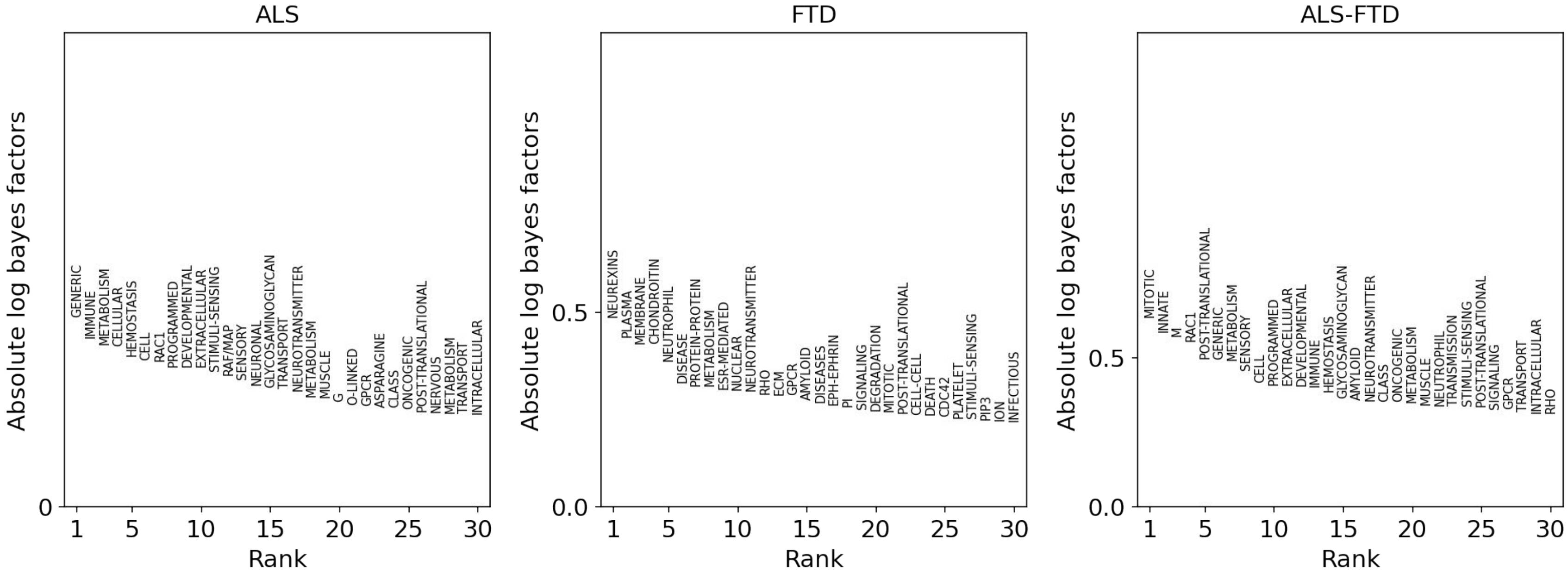
- **Use the scArches/expiMap workflow on user-defined PPI groups**
  - **I have the workflow/code written, I need to establish PPI gene groups**



Planned and future directions

- Follow up on cell-type specific interactions (i.e. heatmap clusters) and see whether they're modulated by in different contexts
- Use the scArches/expiMap workflow on user-defined PPI groups
  - I have the workflow/code written, I need to establish PPI gene groups
- Finish pseudobulked DGE analysis for the ALS/MC1 data
- Elaborate on interesting novel PPIs
- Improve visualization (reducing granularity of neuronal subtypes might help visualization; especially given a tacit focus on glia)
- Integrate spatial data to better inform patterns of CCI and filter less-likely relationships